

ECON 105, Fall 2025
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Logarithm Expansion Practice

Expand each expression using the rules of natural logarithms.

1. $\ln\left(\frac{x^2y^3}{(x+y)^2}\right)$

2. $\ln\left(\frac{(3ex^4y)^2}{\sqrt{x^2+y^2}}\right)$

3. $\ln\left(\frac{(x+2ey)^3}{x^2y^5}\right)$

4. $\ln\left(\sqrt[3]{\frac{e^xy^2}{(x-y)^5}}\right)$

5. $\ln\left(\frac{(x^3+y^3)^2}{e^{2x}y}\right)$

6. $\ln\left(\frac{(2x^2e^y)^5}{(x+y)^3}\right)$

$$7. \ln\left(\sqrt{\frac{e^{3x}y^5}{(x-y)^4}}\right)$$

$$8. \ln\left(\frac{x^3(x+ey)^2}{y^4}\right)$$

$$9. \ln\left(\frac{(x^2+4e^y)^3}{x^2y^2}\right)$$

$$10. \ln\left(\frac{(5ex^2y^3)^4}{(x-y)^2}\right)$$

Answer Key — Expanded Natural Logarithms

1. $\ln\left(\frac{x^2y^3}{(x+y)^2}\right)$
 $= \ln(x^2y^3) - \ln((x+y)^2)$
 $= 2\ln x + 3\ln y - 2\ln(x+y)$
2. $\ln\left(\frac{(3ex^4y)^2}{\sqrt{x^2+y^2}}\right)$
 $= 2\ln(3ex^4y) - \frac{1}{2}\ln(x^2+y^2)$
 $= 2(\ln 3 + \ln e + 4\ln x + \ln y) - \frac{1}{2}\ln(x^2+y^2)$
 $= 2 + 2\ln 3 + 8\ln x + 2\ln y - \frac{1}{2}\ln(x^2+y^2)$
3. $\ln\left(\frac{(x+2ey)^3}{x^2y^5}\right)$
 $= 3\ln(x+2ey) - 2\ln x - 5\ln y$
4. $\ln\left(\sqrt[3]{\frac{e^xy^2}{(x-y)^5}}\right)$
 $= \frac{1}{3}[\ln(e^xy^2) - \ln((x-y)^5)]$
 $= \frac{1}{3}(x + 2\ln y - 5\ln(x-y))$
 $= \frac{x}{3} + \frac{2}{3}\ln y - \frac{5}{3}\ln(x-y)$
5. $\ln\left(\frac{(x^3+y^3)^2}{e^{2x}y}\right)$
 $= \ln((x^3+y^3)^2) - \ln(e^{2x}y)$
 $= 2\ln(x^3+y^3) - (2x + \ln y)$
 $= 2\ln(x^3+y^3) - 2x - \ln y$
6. $\ln\left(\frac{(2x^2e^y)^5}{(x+y)^3}\right)$
 $= 5\ln(2x^2e^y) - 3\ln(x+y)$
 $= 5(\ln 2 + 2\ln x + y) - 3\ln(x+y)$
 $= 5\ln 2 + 10\ln x + 5y - 3\ln(x+y)$
7. $\ln\left(\sqrt{\frac{e^{3x}y^5}{(x-y)^4}}\right)$
 $= \frac{1}{2}[\ln(e^{3x}y^5) - \ln((x-y)^4)]$
 $= \frac{1}{2}(3x + 5\ln y - 4\ln(x-y))$
 $= \frac{3x}{2} + \frac{5}{2}\ln y - 2\ln(x-y)$

$$\begin{aligned}
8. \quad & \ln\left(\frac{x^3(x+y)^2}{y^4}\right) \\
&= \ln(x^3) + \ln((x+y)^2) - \ln(y^4) \\
&= 3\ln x + 2\ln(x+y) - 4\ln y
\end{aligned}$$

$$\begin{aligned}
9. \quad & \ln\left(\frac{(x^2 + 4e^y)^3}{x^2y^2}\right) \\
&= 3\ln(x^2 + 4e^y) - 2\ln x - 2\ln y
\end{aligned}$$

$$\begin{aligned}
10. \quad & \ln\left(\frac{(5ex^2y^3)^4}{(x-y)^2}\right) \\
&= 4\ln(5ex^2y^3) - 2\ln(x-y) \\
&= 4(\ln 5 + \ln e + 2\ln x + 3\ln y) - 2\ln(x-y) \\
&= 4\ln 5 + 4 + 8\ln x + 12\ln y - 2\ln(x-y)
\end{aligned}$$